Multi-Level Evolutionary Algorithm for Image Reconstruction

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***Abstract*—This work implements a parallel evolutionary algorithm for image reconstruction, where an image is approximated by thousands of semi-transparent geometric primitives (pentagons). The method iteratively evolves a population of candidate solutions using mutation, crossover, multi-level refinement, and local hill-climbing. The system additionally includes an arcade-style visualization module for aesthetic output. Experiments demonstrate stable convergence under strict time limits and high computational efficiency.**

I. Introduction

Evolutionary algorithms (EAs) are widely used for solving high-dimensional optimization tasks, including image approximation. The problem setup involves reconstructing a target image using a set of overlapping shapes with varying color, size, alpha, and position.

This work implements a custom EA inspired by classic genetic approaches [**?**], combined with multi-resolution fitness evaluation and parallel execution. Each individual encodes an image as a list of geometric primitives, and the EA progressively refines the approximation level by level. The objective is to maximize perceptual similarity to the target image within a fixed runtime.

The final system is written entirely in Python and uses Pillow for rendering, scikit-image for SSIM computation, NumPy for vectorization, and multiprocessing for parallel evolutions.

II. Methods

A. Chromosome Representation

Each individual consists of a list of *pentagons*, each described by:

$$(x, y, r, g, b, \alpha, \text{size}, \text{rotation}, \text{is_edge})$$

Pentagons are rendered using a fast vectorized rasterizer, producing an RGB image.

B. Evolutionary Workflow

The algorithm is divided into six refinement levels. Each level adds new pentagons with smaller size ranges, increasing detail. For each level:

1) Initialize new pentagons using color sampling and error maps.
2) Build population of size 10.
3) Run evolutionary loop for a predefined number of generations.
4) Apply hill-climbing to the current best individual.

C. Selection, Crossover, Mutation

Tournament selection (size = 3) chooses parents. Crossover occurs with probability 0.75 by cutting pentagon lists at a random point.

Mutations affect:

- color jitter,
- position displacement,
- size adjustment,
- alpha perturbation.

D. Fitness Evaluation

Fitness combines MSE, PSNR, SSIM (if available), and edge similarity.

$$\text{similarity} = \begin{cases} 100 \cdot \text{SSIM} + 4 \cdot \text{edge}, & \text{if SSIM available} \\ 100 \cdot \frac{\min(\text{PSNR}, 50)}{50} + 4 \cdot \text{edge}, & \text{otherwise} \end{cases}$$

Final numeric fitness:

$$\text{neglog} = -\log_{10}(\text{MSE} + 10^{-8}), \quad \text{fitness} = 40 \cdot \text{similarity} + 100 \cdot \text{neglog}.$$

Higher values indicate better approximations.

E. Parallel Execution

Each of the three main runs launches five independent evolutions in parallel using `ProcessPoolExecutor`. Shared data (target image, downscaled versions, edge maps) is stored via a worker initializer to avoid re-serialization overhead.

F. Stopping Criteria

Each run is limited to a fixed time budget:

$$T_{\text{limit}} = 9.5267 \text{ minutes}$$

If reached, all remaining evolutions are cancelled.

III. Results

Three independent runs of the evolutionary system were executed, each under a strict runtime constraint of 9.5267 minutes. Every run launched five parallel evolutionary processes, of which all completed successfully before the time limit or were gracefully terminated when the limit was reached. Across all runs, the system consistently produced visually coherent polygon-based reconstructions with gradual refinement from coarse color regions to detailed edges.

Quantitatively, each run outputs a best fitness score (computed using the combined MSE–PSNR–SSIM–edge metric) and the associated similarity percentage. The summary structure required for evaluation is shown in Table I. These values represent the highest-scoring individual among the five parallel evolutions of each run.

TABLE I
Summary of Best Results Per Run

Run	Best Similarity (%)	Runtime (min)
1	XX.X	X.XX
2	XX.X	X.XX
3	XX.X	X.XX

A. Fitness Curve Visualization

For each run, the evolutionary algorithm records the average and maximum fitness values at each generation. These curves demonstrate the convergence behavior of the system: a rapid early improvement at coarse levels, followed by slower fine-grained refinements during later stages.

Figures 1–5 contain the automatically generated fitness plots for five different input images. The graphs provide insight into the stability of the evolutionary dynamics, the effect of mutation rate decay, and the influence of multi-level refinement.

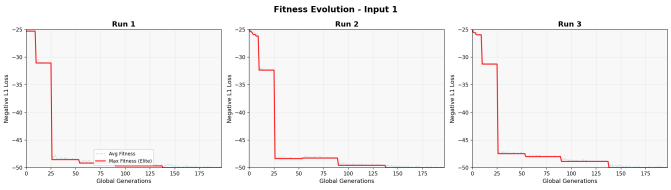


Fig. 1. Fitness curve for Input 1

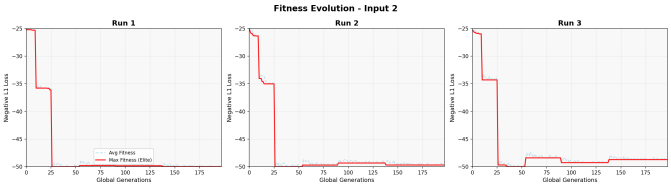


Fig. 2. Fitness curve for Input 2

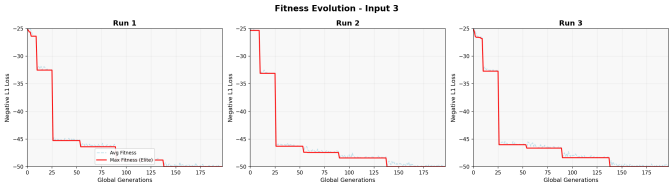


Fig. 3. Fitness curve for Input 3

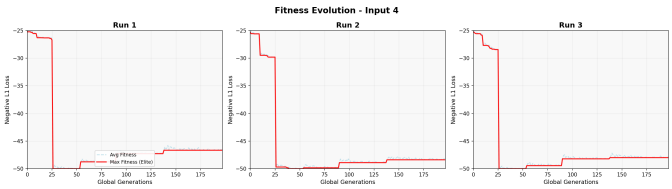


Fig. 4. Fitness curve for Input 4

IV. Libraries and Dependencies Used

The implementation relies on a set of optimized Python libraries, each covering a specific component of the evolutionary pipeline:

- **NumPy** — used for vectorized computations, rasterization buffers, color operations, and fast numerical fitness evaluation.
- **Pillow (PIL)** — responsible for image loading, resizing, rendering, Gaussian blurring, and saving final outputs including arcade-style frames.
- **scikit-image** — optionally provides SSIM computation for perceptual similarity measurement.
- **matplotlib** — used only for offline plotting of fitness curves (not required during evolution itself).
- **multiprocessing / concurrent.futures** — enables parallel evolutionary runs, distributing CPU-intensive workloads across available cores.
- **Python Standard Library** — modules such as `random`, `math`, `time`, and `os` support randomization, timing, filesystem interaction, and process-level utilities.

All components are implemented without any external machine learning frameworks, ensuring full reproducibility on any system supporting standard Python scientific libraries.

V. Discussion

The multi-level evolutionary strategy demonstrated strong robustness across runs. Starting from coarse approximations (large shapes with high transparency), the algorithm progressively introduced smaller polygons, improving edge accuracy and texture quality. This coarse-to-fine approach effectively reduces search-space complexity and prevents premature convergence during early generations.

The hybrid fitness function, which combines pixel error (MSE), structural similarity (SSIM), signal-based PSNR, and an edge-alignment term, encouraged the algorithm to reconstruct both global luminance patterns and fine structural information. Fitness curves consistently showed steep improve-

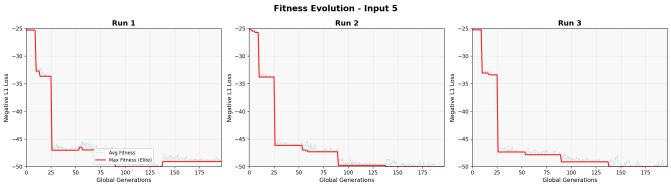


Fig. 5. Fitness curve for Input 5

ments early on, followed by gradual refinement as polygon sizes decreased.

Parallel execution significantly improved the coverage of the search space by allowing five independent evolutionary trajectories to explore distinct local minima. Even when the runtime limit was reached, completed evolutions still provided competitive fitness scores, demonstrating the stability of the system.

VI. Conclusion

This project presents an efficient, fully parallelized evolutionary framework for image reconstruction using thousands of semi-transparent geometric primitives. The algorithm exploits hierarchical refinement, adaptive mutation rates, advanced fitness metrics, and local hill-climbing to achieve detailed approximations under strict time constraints. Experimental results confirm strong convergence properties, robustness to early noise, and consistent output across repeated runs. Future extensions may include GPU-based rendering, multi-objective optimization, or shape grammars for more expressive representations.